

Extra questions (Year 10+ 10A)

Exercise 1.2 Logarithms

1 Work out the following without using a calculator.

- a** $\log 100$ **b** $\log 1$ **c** $\log 0.01$ **d** $\log \sqrt{10}$ **e** $\log \sqrt[4]{10}$ **f** $\log 0.0001$
g $\log \sqrt[5]{10}$ **h** $\log 10^{\frac{5}{2}}$ **i** $\log \sqrt{10^3}$

2 Use a calculator to find the values of the following, and show that they are the solutions of exponential equations.

- a** $\log 72$ **b** $\log 127$ **c** $\log 13$
d $\log 0.03$ **e** $\log 722$ **f** $\log 0.029$
g $\log 32$ **h** $\log 175$ **i** $\log 7.2$

3 Find the values of each of the following.

- a** $\log_3 27$ **b** $\log_6 1296$ **c** $\log_5 625$
d $\log_2 0.0625$ **e** $\log_5 0.0016$ **f** $\log_4 0.0625$
g $\log_2 2048$ **h** $\log_4 128$ **i** $\log_4 \left(\frac{1}{32}\right)$

Use $\log 7.114 = 0.8521$ to find the following:

- a** $\log 71.14$ **b** $\log 7114$ **c** $\log 0.07114$ **d** $\log 0.000\ 7114$

5 **a** Use your calculator to find the logarithm of 488 and use it to find the logarithms of the following.

- b** 48.8 **c** 0.000 488 **d** 4 880 000 **e** 0.0488 **f** 48 800 000 000

6 Given that $\log 3.4 \approx 0.5315$ and $\log 64.5 \approx 1.8096$:

- a** find $\log (3.4 \times 64.5)$ **b** find $\log (64.5 \div 3.4)$

Using the 10^x button on your calculator, hence find:

- c** 3.4×64.5 **d** $64.5 \div 3.4$
e Set out $383.53 \div 99.25$ as it would have been done using logarithms.
f Set out 43.57×53.66 as it would have been done using logarithms.

Answers

- 1** a 2 b 0 c -2 d 0.5 e 0.25 f -4
 g 0.2 h 2.5 i 1.5
2 a $1.8573, 10^{1.8573} \approx 72$ b $2.1038, 10^{2.1038} \approx 127$ c $1.1139, 10^{1.1139} \approx 13$
 d $-1.5229, 10^{-1.5229} \approx 0.03$ e $2.8585, 10^{2.8585} \approx 722$ f $-1.5376, 10^{-1.5376} \approx 0.029$
 g $1.5051, 10^{1.5051} \approx 32$ h $2.2430, 10^{2.2430} \approx 175$ i $0.1761, 10^{0.1761} \approx 1.5$
3 a 3 b 4 c 4 d -4 e -4 f -2
 g 11 h 3.5 i -2.5
4 a 1.8521 b 3.8521 c -1.1479 d -3.1479
5 a 2.6884 b 1.6884 c -3.3116 d 6.6884 e -1.3116 f 10.6884
6 a 2.3411 b 1.2781 c 219.3 d 19.0

e

Number	Logs
383.53	≈ 2.5838
99.25	≈ 1.9967
Subtract for division.	0.5871
Antilog 0.5871 ≈ 3.9	

f

Number	Logs
43.57	≈ 1.6392
53.66	≈ 1.7297
Add for multiplication.	3.3689
Antilog 3.3689 ≈ 2338	